

Assignment 3

Project Name: ARCC (AI Resume and Career Coach)

Course: ENSF 400

Section: 01

Group #: 13

Group Members + UCIDs:

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Team Role:

Ryan Graham: Project Management, Database Control and Cross-Section Support

Fateh Ali: Backend Developer

Rabab Rahman: Frontend Developer

Muhammad Zain: AI/API Integration Developer

Benny Ndabaramiye: Frontend Developer/Backend Support

1. Activity Definition and Sequencing

Activity ID	Activity Name	WBS ID	Description	Duration (Days)
A1	Develop project plan	1.1.1	Define schedule, milestones and iteration structure	1
A2	Create requirements traceability matrix	1.1.2	Map requirements to WBS components and testing	2
A3	Identify project risks and dependencies	1.1.3	Create risks and mitigation strategies	1
A4	Define system requirements and use cases	1.2.1	Translate WBS into backlog and acceptance criteria	3
A5	Design system architecture and components	1.2.2	Define architecture for frontend, backend, database and AI services	3
A6	Create UI/UX framework and navigation	1.2.3	Design layouts for system screens and user flow	2
A7	Implement frontend framework	1.3.1	Configure React structure, routing and shared components	1
A8	Develop resume upload interface	1.3.2	Implement file upload UI with validation and status indicators	1
A9	Develop job description input interface	1.3.3	Build job description submission UI	1

A10	Develop analysis results interface	1.3.4	Create UI for showing resume scores and suggestions	1
A11	Develop interview simulator interface	1.3.5	Implement UI for questions, user inputs, and AI outputs.	2
A12	Implement user dashboard interface	1.3.6	Build dashboard to display user history and their results	1
A13	Implement REST API framework	1.4.1	Build backend Flask project and API endpoints	2
A14	Implement resume parsing module	1.4.2	Extract and normalize resume text	1
A15	Implement resume to job matching engine	1.4.3	Develop simple algorithm for calculating match score (can use AI)	2
A16	Implement skill gap analysis engine	1.4.4	Identify missing skills compared to job description (also connect to AI)	1
A17	Implement resume improvement suggestion engine	1.4.5	Generate resume suggestions with AI	1
A18	Implement interview simulation engine	1.4.6	Generate interview questions and feedback (AI guided by baseline prompts)	2

A19	Integrate speech-to-text	1.4.7	Enable voice response for interview simulation realism	2
A20	Implement user authentication system	1.4.8	Implement login, registration and session handling	1
A21	Design database schema	1.5.1	Create ERD and database schema definitions	1
A22	Implement user data storage	1.5.2	Implement data storage and access for user accounts	1
A23	Implement resume and analysis storage	1.5.3	Store resumes, analysis results and interview data	1
A24	Integrate LLM API service	1.6.1	Implement API calls for suggestions and interview feedback	2
A25	Implement security and encryption	1.6.2	Configure secure credential handling (and ensuring secrets are not leaked)	1
A26	Develop security compliance	1.7.1	Ensure security checks are proper and that sensitive data is only released in proper moments	2
A27	Perform performance optimization	1.7.2	Optimize API responses and database queries	2

A28	Implement reliability and recovery mechanisms	1.7.3	Add retries and error handling	1
A29	Implement scalability considerations	1.7.4	Configure stateless API and resource handling	1
A30	Implement accessibility improvements	1.7.5	Ensure UI accessibility compliance	2

2. Network Diagram and Critical Path

The critical path is highlighted in red:

